Yuzhang Xie

4th-Year Ph.D. Candidate
Department of Computer Science
Emory University Atlanta, GA, USA

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Research Interests

Multimodal AI for Healthcare: Developing robust and interpretable AI methods that integrate diverse clinical data modalities, including tables, text, images, omics, and time-series data, to support clinical decision-making.
LLM Agents & Reliable Learning: Studying how LLM-based agents improve through interaction, fine-tuning, and feedback, and how to evaluate their reasoning and decisions reliably.
KG-Guided Learning & Reasoning: Integrating biomedical knowledge graphs into machine learning, especially LLMs, to support grounded reasoning, inference, and interpretability.
Data Mining & Representation Learning: Extracting knowledge from heterogeneous data and learning transferable representations with language models and graph neural networks.

Education

Emory University — **Atlanta, GA, USA**
Ph.D. in Computer Science and Informatics (GPA: 3.98/4.00) — *Aug. 2023 – May 2028 (expected)*
- Emory Graph Mining Lab & Data Science Center of Emory Nursing School
- Advisors: Dr. Carl J. Yang & Dr. Xiao Hu

Shanghai Jiao Tong University (SJTU) — **Shanghai, China**
M.S. in Computer Science (GPA: 3.88/4.00, Rank: 7/126, Top 5%) — *Sept. 2020 – Mar. 2023*
- SJTU Center for Brain-like Computing and Machine Intelligence
- Advisor: Dr. Yang Yang

Shanghai Jiao Tong University (SJTU) — **Shanghai, China**
B.S. in Environmental Science and Engineering (GPA: 3.72/4.00, Rank: 16/59, Top 30%) — *Sept. 2016 – June 2020*

Industry Experience

GE HealthCare — **Bellevue, WA, USA**
AI Scientist Intern — *May 2025 – Aug. 2025*
- Designed scalable patient-level pipelines that integrate heterogeneous clinical modalities to support generalizable foundation models for clinical prediction.
- Fine-tuned language models for biomedical clinical coding, improving the mapping of free-text notes to standardized medical codes.
- Mentors: Dr. Andrew Warrington & Dr. Xiaojian Xu

Research Experience

Emory University — **Atlanta, GA, USA**
Research Assistant — *Aug. 2023 – Present*
- Developing multimodal clinical prediction models for diagnosis, prognosis, and treatment planning using heterogeneous patient data.
- Building and evaluating multi-agent LLM approaches for clinical reasoning, including knowledge grounding, iterative feedback, and self-refinement.
- Designing automated benchmarks and reliable evaluation methods for LLM decision-making in clinical tasks and simulated interactions.
- Applying knowledge graph-guided learning to improve the accuracy and interpretability of clinical predictions.

Shanghai Jiao Tong University — **Shanghai, China**
Research Assistant — *Sept. 2020 – Mar. 2023*
- Developed multimodal learning pipelines integrating heterogeneous clinical data to improve cancer diagnosis and prediction.
- Conducted model interpretability analyses based on feature attribution to identify key predictive factors and assess their contributions to cancer-related risks.

Publications

First-Author Publications

[KDD 2026] **Yuzhang Xie**, Keqi Han, Yunpeng Xiao, Hejie Cui, Guanchen Wu, Ziyang Zhang, Kai Shu, Jiaying Lu, Xiao Hu, Carl Yang. "EHRBench: An Automated and Reliable EHR-based Benchmark for Clinical Decision Making with LLMs." *32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2026, 10068–10079. **Oral**.

[AMIA 2026] **Yuzhang Xie**, Yuhua Wu, Ruiyu Wang, Fadi Nahab, Xiao Hu, Carl Yang. "Enhanced Atrial Fibrillation Prediction in ESUS Patients with Hypergraph-based Pre-training." *American Medical Informatics Association Summits on Translational Science Proceedings*, 2026, 526–535. **Oral**.

[AMIA 2025] **Yuzhang Xie**, Hejie Cui, Ziyang Zhang, Jiaying Lu, Kai Shu, Fadi Nahab, Xiao Hu, Carl Yang. "KERAP: A Knowledge-Enhanced Reasoning Approach for Accurate Zero-shot Diagnosis Prediction Using Multi-agent LLMs." *American Medical Informatics Association Annual Symposium*, 2025. **Oral**.

[ISWC 2025] **Yuzhang Xie**, Xu Han, Ran Xu, Xiao Hu, Jiaying Lu, Carl Yang. "HypKG: Hypergraph-based Knowledge Graph Contextualization for Precision Healthcare." *24th International Semantic Web Conference, Main Conference, Research Track*, 2025. **Oral**.

[SIGIR 2024] **Yuzhang Xie**, Jiaying Lu, Joyce Ho, Fadi Nahab, Xiao Hu, Carl Yang. "PromptLink: Leveraging Large Language Models for Cross-Source Biomedical Concept Linking." *47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024.

[AI in Med.] **Yuzhang Xie**, Qingqing Sang, Qian Da, Guoshuai Niu, Shijie Deng, Haoran Feng, et al. "Improving diagnosis and outcome prediction of gastric cancer via multimodal learning using whole slide pathological images and gene expression." *Artificial Intelligence in Medicine*, 152 (2024): 102871.

[BIBM 2022] **Yuzhang Xie**, Guoshuai Niu, Qian Da, Wentao Dai, Yang Yang. "Survival Prediction for Gastric Cancer via Multimodal Learning of Whole Slide Images and Gene Expression." *IEEE International Conference on Bioinformatics and Biomedicine*, 2022. **Oral**.

Co-Authored Publications

[EMNLP 2026] Mingyang Wei, Dehai Min, Zewen Liu, **Yuzhang Xie**, Guanchen Wu, Ziyang Zhang, Carl Yang, Max S. Y. Lau, Qi He, Lu Cheng, Wei Jin. "EpiQAL: Benchmarking Large Language Models in Epidemiological Question Answering and Reasoning." *Accepted to the Conference on Empirical Methods in Natural Language Processing*, 2026.

[AMIA 2026] Mingchen Shao, **Yuzhang Xie**, Carl Yang, Jiaying Lu. "LLM-MINE: Large Language Model based Alzheimer's Disease and Related Dementias Phenotypes Mining from Clinical Notes." *Accepted to the American Medical Informatics Association Annual Symposium*, 2026.

[BHI 2026] Guanchen Wu, Zhe Huang, **Yuzhang Xie**, Runze Yan, Akul Chopra, Deqiang Qiu, Xiao Hu, Fei Wang, Carl Yang. "MIRAGE: Knowledge Graph-Guided Cross-Cohort MRI Synthesis for Alzheimer's Disease Prediction." *Accepted to IEEE-EMBS International Conference on Biomedical and Health Informatics*, 2026.

[AACL 2026] Yunpeng Xiao, **Yuzhang Xie**, Junhao Huang, Carl Yang, Xiao Hu, Kai Shu. "Decisions by Discrimination: Differential Evidence for Medical Reasoning." *Accepted to the 5th Asia-Pacific Chapter of the Association for Computational Linguistics*, 2026.

[BIBM 2025] Guanchen Wu, **Yuzhang Xie**, Huanwei Wu, Zhe He, Hui Shao, Xiao Hu, Carl Yang. "Utilizing Large Language Models for Zero-Shot Medical Ontology Extension from Clinical Notes." *IEEE International Conference on Bioinformatics and Biomedicine*, 2025.

[Agents4Science 2025] Guanchen Wu, Zuhui Chen, **Yuzhang Xie**, Carl Yang. "Towards Automatic Evaluation and Selection of PHI De-identification Models via Multi-Agent Collaboration." *Open Conference of AI Agents for Science, Spotlight*, 2025.

[ACL 2025] Meng Lu, **Yuzhang Xie**, Zhenyu Bi, Shuxiang Cao, Xuan Wang. "CROSSAGENTIE: Cross-Type and Cross-Task Multi-Agent LLM Collaboration for Zero-Shot Information Extraction." *Findings of the Association for Computational Linguistics*, 2025.

[NPJ Dig. Med.] Balu Bhasuran, Qiao Jin, **Yuzhang Xie**, Carl Yang, Karim Hanna, Jennifer Costa, Cindy Shavor, Zhiyong Lu, Zhe He. "Preliminary analysis of the impact of lab results on large language model generated differential diagnoses." *NPJ Digital Medicine*, 8.1 (2025): 166.

[KDD 2024] Ziyang Zhang, Hejie Cui, Ran Xu, **Yuzhang Xie**, Joyce Ho, Carl Yang. "TACCO: Task-guided Co-clustering of Clinical Concepts and Patient Visits for Disease Subtyping based on EHR Data." *30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2024.

Honors & Awards

- SIGIR 2024 NSF Travel Award — Jul. 2024
- Emory University Laney Graduate Student Council Travel/Research Grant — Jul. 2024
- Outstanding Graduates Award, Shanghai Jiao Tong University — Mar. 2023
- Academic Excellence Scholarship (Top 30%), Shanghai Jiao Tong University — 2017, 2018, 2019, 2021, 2022
- Outstanding Teaching Assistant, Shanghai Jiao Tong University — Mar. 2022
- Outstanding Student, Shanghai Jiao Tong University — May 2019
- Outstanding Leader of Student Organization, Shanghai Jiao Tong University — May 2017

Services

Teaching Assistant

- CS526 Algorithms, Emory University — Fall 2024
- CS524 Theory of Computing, Emory University — Spring 2024
- CS325 Artificial Intelligence, Emory University — Fall 2023
- CS7336 Machine Learning, Shanghai Jiao Tong University — Spring 2022
- CS3317 Artificial Intelligence, Shanghai Jiao Tong University — Fall 2021

Student Organization

- Vice President, Volunteer Guide Association, Shanghai Jiao Tong University — Mar. 2017 – Mar. 2018

Volunteer

- Conference Volunteer, SIGIR — Jul. 2024
- Volunteer Guide for Campus Guests, Shanghai Jiao Tong University — 2016 – 2018

Conference & Journal Reviewer

- ACM Computing Surveys — 2026
- Nutrition Reviews — 2026
- BMC Medical Education — 2026
- Discover Applied Sciences — 2026
- SIAM International Conference on Data Mining (SDM) — 2026
- European Heart Journal — 2026
- BMC Bioinformatics — 2026
- IEEE Transactions on Medical Imaging (TMI) — 2025
- Annual Meeting of the Association for Computational Linguistics (ACL) — 2025
- Journal of Biomedical Informatics (JBI) — 2025
- International World Wide Web Conference (WWW) — 2024, 2025
- ACM Conference on Research and Development in Information Retrieval (SIGIR) — 2024, 2025
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS) — 2023, 2024
- International Conference on Information and Knowledge Management (CIKM) — 2024
- Big Data — 2024

Skills

- LLMs & Agents: Fine-tuning, multi-agent systems, knowledge-grounded reasoning, iterative refinement, benchmark design, evaluator reliability
- Machine Learning: Multimodal learning, representation learning, graph neural networks, knowledge graph-guided reasoning
- Data Modalities: Tables, texts, images, omics, longitudinal time series
- Programming Languages & Tools: Python, LaTeX, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, Pyhealth, OpenSlide, Wandb